Structure Prediction and Comparative Analysis of the Human μ-Opioid Receptor

Question 1

1.2

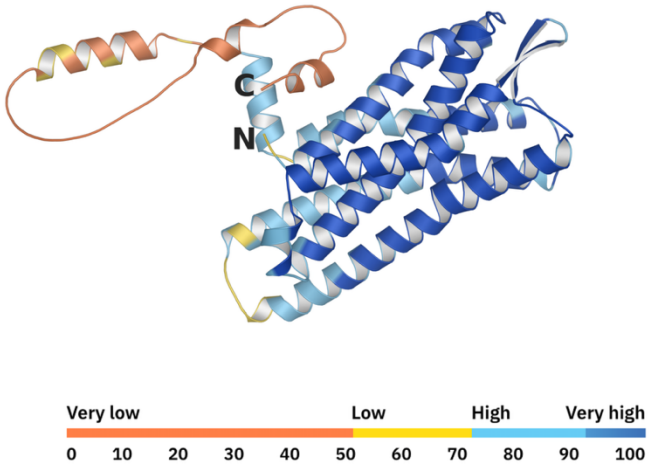


*Figure 1. Predicted structures of MOR from AlphaFold, colored by per-residue measure of local confidence. Model confidence is measured by **pLDDT** (predicted Local Distance Difference Test) on a 0 – 100 scale. Higher scores indicate regions of reliable structural accuracy, whereas low-scoring regions are structurally uncertain.*

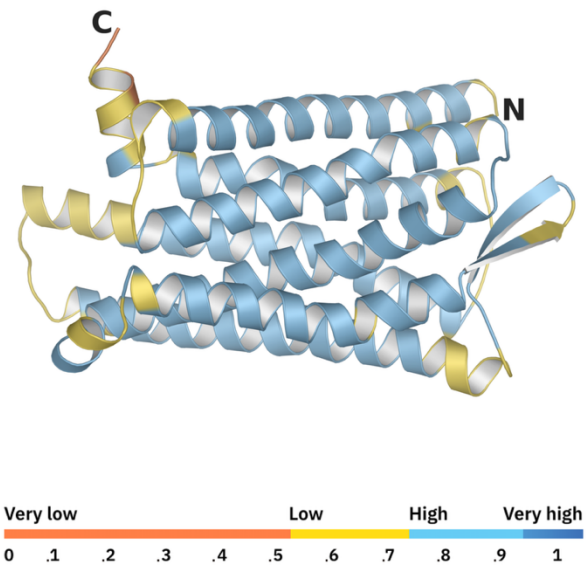


Figure 2. Predicted structures of MOR from AlphaFold, colored by per-residue measure of local confidence. Model confidence is measured by ***QMEANDisCo*** *(Qualitative Model Energy ANalysis with Distance Constraints) on a 0 – 1 scale. Higher scores indicate regions of reliable structural accuracy, whereas low-scoring regions are structurally uncertain.*

Lowest Confidence Regions:

- For AlphaFold, the regions with the lowest confidence (colored in orange) are located in **C-termini and adjacent cytoplasmic regions** (intracellular loops). **N-termini and extracellular loop 2** (ECL2, between TM4 and TM5) are also in low confidence (colored in yellow).
- For SWISS-MODEL, the regions with the lowest confidence (colored in orange) are located in the **C-termini**. **N-termini and several intracellular & extracellular loops** are also in low confidence (colored in yellow).

Biochemical Reasons:

- For N/C-termini, they are frequently i**ntrinsically disordered without a specific ligand**. Also, they are often subject to **post-translational modifications**. Thus, it is challenging for them to achieve a singular, stable conformation, which in turn diminishes both predictive accuracy and experimental clarity.
- For ICLs/ECLs, some of them function as the contact interface of the MOR with G-proteins or ligands. These segments involve **remodeling in response to signals**, resulting in conformational changes of the receptor. Hence, no singular stable conformation is conclusive for these loops.
- Moreover, considering the evolution factors, both SWISS-MODEL and AlphaFold partially depend on evolutionary data. **Loop regions exhibit weak conservation in evolution**, resulting in restricted co-evolutionary couplings and variable templates, hence degrading prediction precision.

1.3

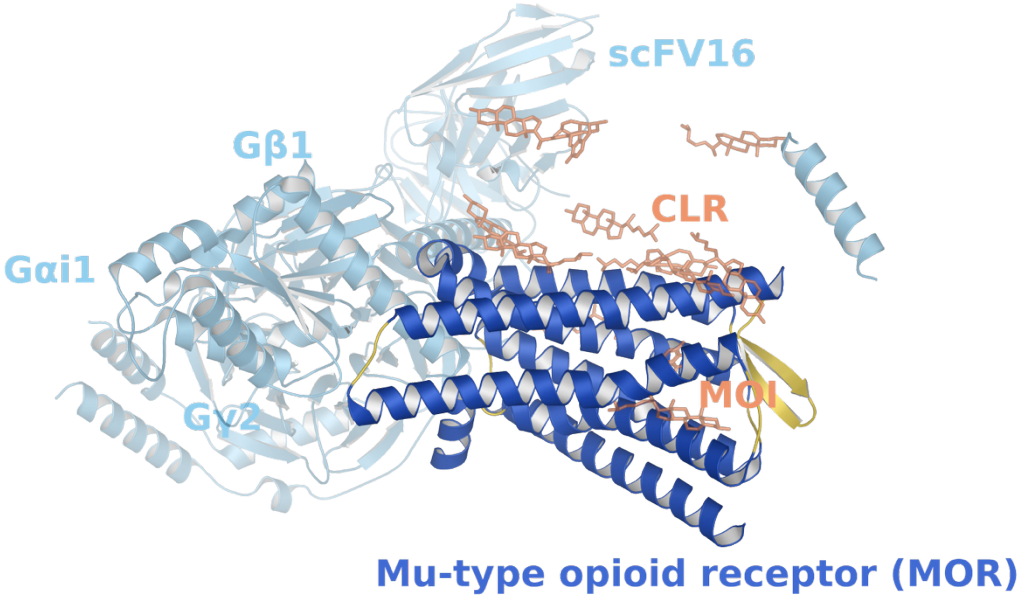


Figure 3. PDB 8EF6: Morphine-bound mu-opioid receptor-Gi complex. The seven transmembrane helices of MOR are colored in deep blue cartoons; ICLs/ECLs are shown in yellow. Co-complexed components (ligands in orange, G-proteins in light blue) are also included in the PDB to ***immobilize MOR into a singular, fixed conformation, thereby enhancing cryo-EM resolution & quality****. For example, MOI and certain G-proteins induce and stabilize the active conformation of the receptor; scFV16 secures the assembly; CLR mimics the native bilayer environment.*

Question 2

2.1

	RMSD	*TM-Score*
AlphaFold	***1.46***	***0.96***
SWISS-MODEL	***0.4***	***1***

Table 1. RMSD and TM-Score for the alignment of AlphaFold and SWISS-MODEL models against the experimental structure.

Overall Fold Quality:

- **SWISS-MODEL is the better** fit to the experimental structure.
- The model by SWISS-MODEL has a lower RMSD, indicating a closer overall atomic position to the experimental structure compared to the model by AlphaFold.
- Additionally, SWISS-MODEL prediction has a higher TM-Score (1, i.e., an identical fold), demonstrating a more similar fold structure to experimental data.

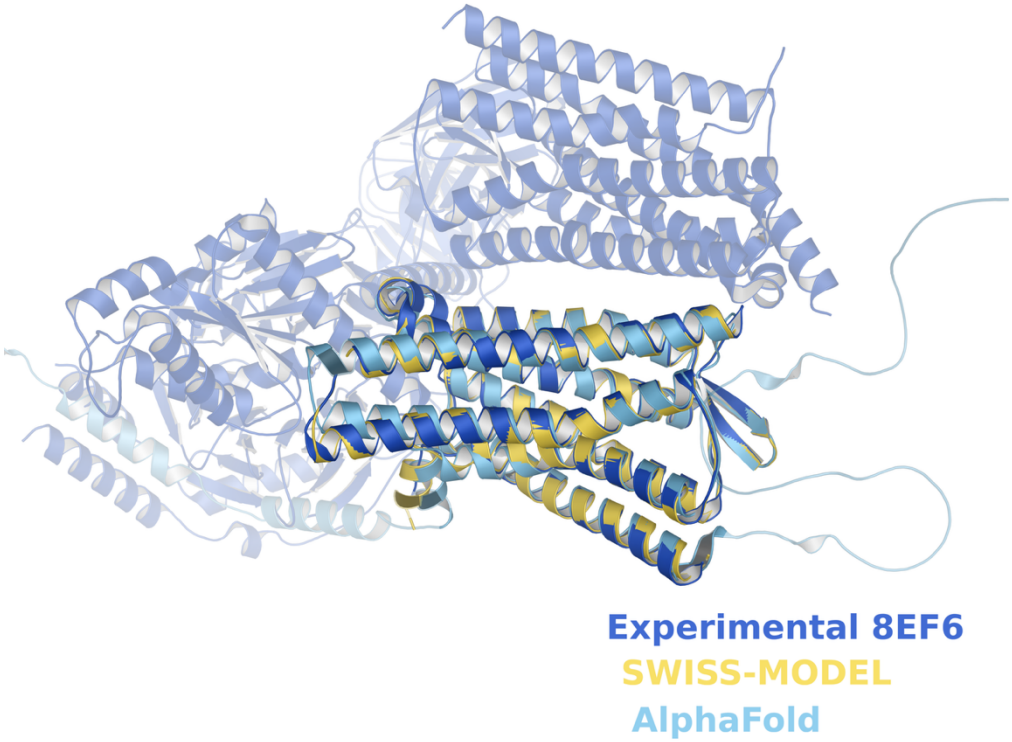


Figure 4. Superposition of three MOR structures. Experimental 8EF6 in dark blue, SWISS-MODEL in yellow, and AlphaFold in light blue.

2.2

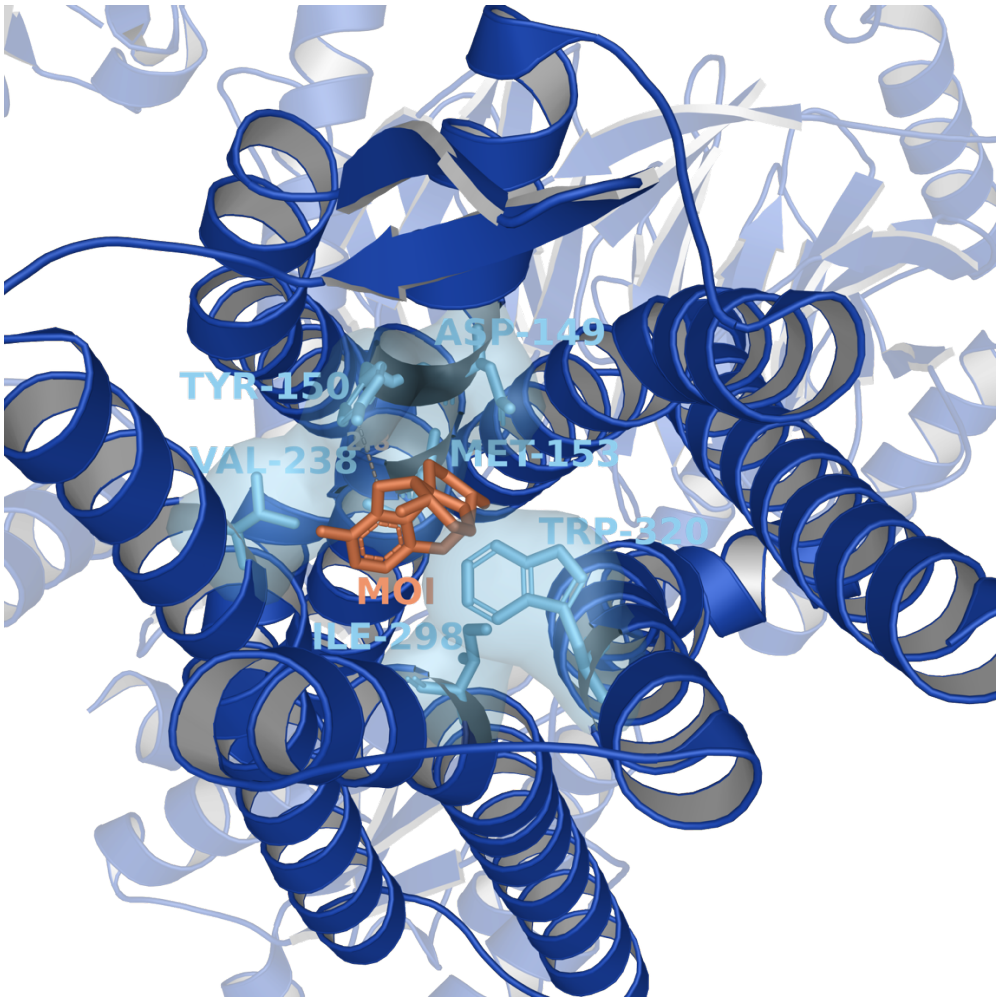


Figure 5. MOR orthosteric pocket in 8EF6. The agonist MOI is drawn as orange sticks. Key interacting residues are shown in light blue sticks (a possible interaction between atoms within 4 Å is also denoted in a grey dash).

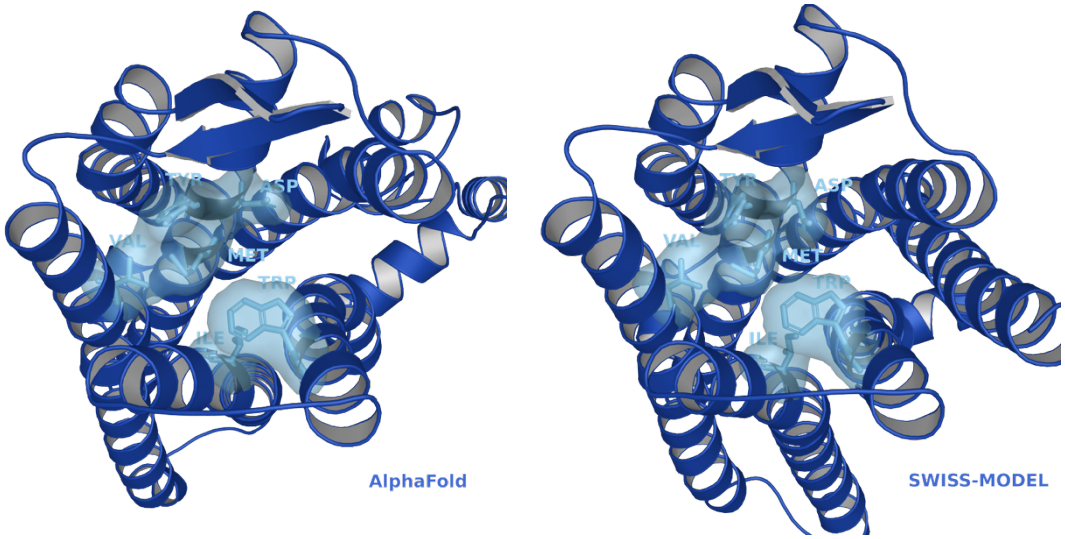


Figure 6. MOR binding pockets in models from AlphaFold and SWISS-MODEL. Key interacting residues are shown in light blue sticks.

Similarities And Differences:

- Both predicted pockets accurately replicate the experimental residue positions and orientations, although slight differences exist.
- **Backbone:** The SWISS-MODEL closely aligns with the actual conformation, while the AlphaFold model exhibits **a slight angular variance in part of TM6**.
- **Side chain:** SWISS-MODEL closely matches the experimental conformation, whereas in AlphaFold, the ASP side chain shows a **distinct orientation**; and the aromatic rings of Try and MET exhibit minor **rotameric changes**; also, the **relative positions** among these residues in the binding pocket are somewhat altered compared to the experimental configuration, reducing the TM3–TM6 opening.

2.3

- For SWISS-MODEL:
  - Backbone heavy-atom **RMSD** is small: ICL3 > ICL1 > ICL2 (Table 2).
  - All three ICLs are broadly structurally similar to the active conformation with only minor positional deviations, maintaining an **extended, cytoplasm-facing trajectory for G-protein interaction**.
- For AlphaFold:
  - Backbone heavy-atom **RMSD** is large: ICL3 > ICL1 > ICL2 (Table 2).
  - ICL1/2 deviations are mainly **local (on loops themselves)** and do not significantly disrupt the TM1–TM2 or TM3-TM4 orientations.
  - In contrast, ICL3 exhibits low similarity to the experiment configuration: **the loop is shorter and inward-oriented**, resulting in a **slight inward angular displacement of TM6 and a reduction in the cytoplasmic opening of MOR**. This reduces the accuracy and confidence of the ultimate prediction of the active state of MOR.

Differences Between Intracellular Loops:

	ICL1 (94–106)	***ICL2 (166–185)***	***ICL3 (256–279)***
SWISS-MODEL	***0.462***	***0.425***	***0.548***
AlphaFold	***1.149***	***1.134***	***3.705***

Table 2. RMSD for SWISS-MODEL & AlphaFold models to the active-state reference (8EF6) in ICLs. Only heavy atoms in backbones are measured. (Unit: Å)

Question 3

3.1

Success Of Prediction Methods:

- **SWISS-MODEL** produces a more relevant model of MOR.
- Globally, it aligns more closely with the experimental fold (higher **TM-Score** and lower **RMSD** for the superimposed atoms).
- Locally, the **ICL** predictions exhibit superior alignment with the experimental active state fold, and the **binding pocket** demonstrates more reasonable residue placements and orientations.

Reasons Based On Methods Principles:

- **Homology Modeling (SWISS-MODEL)**
  - Homology modeling uses templates with 3D structures determined by empirical methods. When conducting prediction, we choose “8ef5.1.F”, a fentanyl-bound active MOR structure as a template due to its **high sequence identity, extensive coverage, and high-quality experimental measurements**.
  - Moreover, **the small-molecule agonist in this template (fentanyl) acts similarly to morphine**. Thus, this template can better simulate the stable small-molecule-agonist-activated state of MOR with a more reasonable binding pocket.
- **Machine Learning (AlphaFold)**
  - In comparison, AlphaFold utilizes advanced machine learning techniques to predict protein structures. However, its weakness in our prediction might exist in three aspects listed below:
  - Firstly, the prediction process involves **coevolutionary information**. However, N/C-termini or loop regions exhibit weak conservation in evolution, resulting in less convincing MSA results and variable templates, which degrade prediction precision.
  - Secondly, an **end-to-end approach** is another feature for AlphaFold prediction. As the active state of MOR may involve some configuration changes as intermediate steps or templates, AlphaFold may ignore the variance and produce a less acceptable result.
  - Thirdly, AlphaFold was trained on proteins with structures resolved in the PDB. It is **not intrinsically designed to predict the flexibility or structures of flexible regions**, such as ICLs/ECLs, although the per-residue confidence scores are clearly presented.

3.2

Biological States:

- For **SWISS-MODEL**, it is more likely to represent an active state of MOR.
  - **Evidence 1 – Intracellular face:** ICL3 in SWISS-MODEL exhibits a strong alignment with the active reference (RMSD = 0.548 Å), maintaining an **extended, cytoplasm-facing trajectory for G-protein interaction**.
  - **Evidence 2 – Transmembrane core:** SWISS-MODEL replicates the displacement of TM6 superposed to 8EF6, which is a characteristic of activation.

- For **AlphaFold**, it is more likely to represent an inactive state of MOR.
  - **Evidence 1 – Intracellular face:** ICL3 in AlphaFold weakly aligned to 8EF6 (RMSD = 3.705 Å). **The loop is shorter and inward-oriented, indicating a narrower cytoplasmic cavity.**
  - **Evidence 2 – Transmembrane core:** AlphaFold demonstrates **a slight inward displacement of TM6, reducing the TM3–TM6 opening**, representing a potential inactive state of MOR.